

RAPID CITY, SD, USA

WMO: 726620

Lat: 44.046N

Lon: 103.054W

Elev: 963

StdP: 90.28

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24090

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-19.4	-26.0	0.4	-22.2	-23.0	0.5	-18.9	18.5	-1.9	16.8	-2.2	3.3	350	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	35.9	18.9	33.6	18.8	31.6	18.5	21.9	29.6	20.9	29.0	20.0	28.4	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.0	25.9	18.4	14.9	24.7	17.3	13.9	23.6	68.7	29.7	64.9	29.0	61.5	28.1	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.7	11.7	-27.0	39.0	3.0	2.5	-29.1	40.8	-30.9	42.2	-32.6	43.6	-34.7	45.4		
(5)			-27.1	23.6	2.6	1.2	-29.0	24.5	-30.5	25.2	-32.0	25.9	-33.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-3.9	-3.1	1.9	6.9	12.4	18.4	23.0	21.9	16.7	8.8	1.9	-3.5
	DBStd	11.19	7.40	7.80	7.21	5.57	4.72	4.16	3.50	3.43	5.02	5.50	6.52	7.11
	HDD10.0	1974	432	369	259	122	29	1	0	0	7	86	249	419
	HDD18.3	3952	691	601	508	344	190	48	4	9	89	297	493	677
	CDD10.0	1431	0	1	9	28	104	253	402	370	207	50	6	0
	CDD18.3	367	0	0	0	1	7	50	148	121	38	2	0	0
	CDH23.3	4751	0	0	9	39	133	661	1784	1432	614	76	3	0
	CDH26.7	2150	0	0	1	7	33	259	889	679	261	21	0	0
Wind	WSAvg	4.6	4.5	4.8	5.0	5.5	5.1	4.4	4.1	4.0	4.3	4.8	4.5	4.3
Precipitation	PrecAvg	450	9	10	23	55	89	73	55	47	32	37	12	8
	PrecMax	601	26	23	48	122	186	161	99	107	76	136	33	27
	PrecMin	310	0	0	1	10	38	17	15	20	5	2	1	0
	PrecStd	88	7	7	13	26	44	40	24	22	18	32	9	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	19.5	24.5	27.6	30.3	36.6	39.7	37.6	35.7	29.2	22.9	16.2
		MCWB	6.4	8.4	10.5	12.8	17.2	18.4	19.4	19.0	17.7	14.2	10.0	7.2
	2%	DB	12.0	14.6	20.2	23.8	26.9	32.4	36.4	35.2	32.6	25.3	18.2	11.9
		MCWB	4.9	6.1	9.1	12.0	16.0	19.0	19.5	18.6	16.8	12.9	8.1	4.8
	5%	DB	9.0	10.9	16.7	20.2	24.1	29.5	33.9	32.9	29.7	21.7	15.1	8.8
		MCWB	3.3	4.4	7.8	10.5	15.0	18.8	19.5	18.6	16.3	11.3	6.9	3.0
	10%	DB	6.3	7.5	13.2	17.0	21.7	27.1	31.6	30.6	27.0	18.3	11.7	5.9
		MCWB	1.8	2.4	6.0	9.0	13.8	17.8	19.1	18.6	15.3	10.0	5.3	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.7	11.4	14.2	19.1	22.8	23.6	22.7	20.3	15.6	10.5	7.4
		MCDB	15.0	18.2	22.1	24.0	26.9	29.0	31.1	30.2	30.0	25.5	21.7	15.5
	2%	WB	5.2	6.4	9.6	12.4	17.2	21.1	22.2	21.2	18.7	13.6	8.6	5.1
		MCDB	11.4	14.0	19.3	22.0	24.6	28.1	30.1	29.4	28.2	23.0	17.4	11.4
	5%	WB	3.6	4.6	8.1	11.0	15.7	19.9	21.3	20.3	17.4	12.0	7.0	3.3
		MCDB	8.5	10.3	16.2	19.3	22.8	27.6	29.5	28.7	27.2	20.0	14.5	8.4
	10%	WB	2.0	2.7	6.3	9.5	14.2	18.7	20.3	19.4	16.1	10.5	5.4	1.7
		MCDB	6.0	7.2	12.8	16.2	20.4	26.0	28.8	27.7	25.2	17.5	11.4	5.8
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.5	13.4	13.2	12.9	13.7	14.9	15.3	15.4	13.9	13.2	11.9
		MCDBR	16.1	17.5	19.7	19.8	17.2	18.0	18.5	18.8	19.8	19.8	18.5	15.6
		MCWBR	10.2	10.4	10.6	10.2	8.8	8.0	6.6	6.9	8.0	9.7	10.3	9.8
	5% WB	MCDBR	15.4	16.6	18.6	18.0	16.0	16.0	15.3	15.7	17.6	17.8	17.5	15.1
		MCWBR	10.1	10.1	10.3	9.7	8.6	8.4	7.0	6.8	7.6	9.0	10.0	9.7
Clear-Sky Solar Irradiance	taub		0.253	0.262	0.280	0.315	0.333	0.346	0.369	0.364	0.327	0.290	0.263	0.252
	taud		2.577	2.551	2.548	2.436	2.462	2.451	2.419	2.415	2.499	2.577	2.564	2.563
	Ebn at Noon		906	953	970	950	937	921	895	889	902	904	882	872
	Edh at Noon		63	77	88	108	108	110	112	109	92	74	63	58
All-Sky Solar Radiation	RadAvg		1.93	2.89	4.05	5.03	5.97	6.94	7.04	6.06	4.79	3.15	2.13	1.61
	RadStd		0.10	0.17	0.31	0.30	0.57	0.51	0.26	0.27	0.29	0.30	0.16	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page